Doc. 188 · FFGFT Research Series

Geometric Foundations of FFGFT

From the Phase Condition to Circle, Polygon, Spiral and the Preferred Ratios of Nature

Information

Abstract. The Fundamental Fractal Geometric Field Theory (FFGFT) derives all stable geometric forms from a single condition: phase coherence after a complete torus orbit. From this condition follow, in strict hierarchy, regular polygons, the circle as a limiting case, the spiral as superposition of two frequencies, the golden ratio ϕ as the most stable incommensurable ratio, the Fibonacci sequence as its integer approximation, and all derived FFGFT formulae — lepton scale ladder, $1/n^2$ spectrum, fractal coupling constant $\alpha = \phi^{-1}$. The same hierarchy appears in crystal lattices, spiral galaxies, biological growth patterns, and quantum numbers.

Cross-references: Doc. 180 (Lagrangian) · Doc. 182 (Universal scale) · Doc. 187 (TFLN photonics) · GitHub: jpascher/T0-Time-Mass-Duality · Zenodo: DOI 10.5281/zenodo.18834145

Contents

1 The Universal Phase Condition

The starting point of the entire geometric hierarchy is a single condition. A field configuration Ψ can only exist stably if it agrees exactly with itself after one complete orbit on the 4D torus:

$$\oint_{\mathcal{T}^4} d\phi = 2\pi \cdot n, \quad n \in \mathbb{N}. \tag{1}$$

Fundamental Principle

Equation (1) is the only selection rule of FFGFT. All stable structures — geometric forms, particle masses, constants of nature, preferred ratios — follow from this condition. Not as approximations, but as exact mathematical consequences.

The phase condition admits three solution classes:

1. **Discrete solutions:** $n \in \{1, 2, 3, ...\}$ — regular polygons, quantum numbers, crystal symmetries.
2. **Continuous limiting cases:** $n \to \infty$ — circle, ellipse, smooth curves.
3. **Simultaneous solutions for two frequencies:** leads to incommensurable ratios — spiral, golden ratio, Fibonacci sequence.

2 Level 1: Discrete Symmetries — Polygons and Crystals

2.1 Regular Polygons

For integer n in Eq. (1), a regular n-polygon emerges as the geometric realisation of the phase condition. Vertices at:

$$\theta_k = \frac{2\pi k}{n}, \quad k = 0, 1, \ldots, n-1. \tag{2}$$

2.2 Special Role of $n = 3$: The Z3 Symmetry

The triangle ($n = 3$) is the most economical stable discrete solution — three points are the minimum for a closed non-trivial configuration. In FFGFT, the three particle generations have exactly this Z3 structure:

$$\mathbb{Z}_3: \quad \theta \mapsto \theta + \frac{2\pi}{3}, \qquad \text{Generation } g \in \{1, 2, 3\} \;\longleftrightarrow\; \theta_g = \frac{2\pi(g-1)}{3}. \tag{3}$$

2.3 $n = 6$: Hexagonal Packing and Crystal Lattices

$$\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2. \tag{4}$$

Honeycomb structure, graphene lattice, lithium niobate crystal lattice — all realise this symmetry at different scales.

2.4 Quantum Numbers as Polygon Orders

The principal quantum number n corresponds directly to the polygon order:

$$E_n = -\frac{E_0}{n^2}, \quad n = 1, 2, 3, \ldots \tag{5}$$

The $1/n^2$ spectrum is the energy distribution across stable polygon modes — not a coincidence, but a consequence of Eq. (1).

3 Level 2: Circle and Ellipse

3.1 Limiting Case $n \to \infty$: Continuous Phase Densification

$$\lim_{n\to\infty} \left\{ \frac{2\pi k}{n} \,\middle|\, k = 0, \ldots, n-1 \right\} = [0, 2\pi). \tag{6}$$

The circle is not geometrically fundamental — it is the limiting case of an infinite sequence of polygons.

3.2 Ellipse: Broken Symmetry

When the phase condition is satisfied in two spatial directions with different coupling strengths:

$$\oint_x d\phi = 2\pi n_x, \qquad \oint_y d\phi = 2\pi n_y, \quad n_x \neq n_y, \tag{7}$$

an ellipse with axis ratio $n_x : n_y$ emerges. Planetary orbits, higher angular momentum orbitals, resonator modes — all follow this logic.

4 Level 3: Spirals — Superposition of Two Frequencies

4.1 Origin of the Spiral

A spiral arises when two phase conditions must be satisfied simultaneously. For rational ratio $\omega_r/\omega_\theta = p/q$ the curve closes after q orbits. For irrational ratios it never closes — a true spiral emerges:

$$r(\theta) = r_0 \cdot e^{(\omega_r/\omega_\theta)\,\theta}. \tag{8}$$

4.2 The Logarithmic Spiral as Self-Similarity Solution

$$r(\theta) = a \cdot b^\theta, \qquad r(\theta + 2\pi) = b^{2\pi} \cdot r(\theta). \tag{9}$$

This is the only curve that agrees with itself under scaling — the geometric realisation of fractal self-similarity. In nature: nautilus spiral, galaxy arms, phyllotaxis, hurricane structure.

5 Level 4: The Golden Ratio ϕ

5.1 Two Simultaneous Phase Conditions

Which irrational number is most stable against destructive interference? The one that is most *poorly* approximable by rationals:

$$\phi = \frac{1+\sqrt{5}}{2} = 1.6180339\ldots \tag{10}$$

Why ϕ is Optimal

ϕ has the continued fraction representation with all ones — the smallest possible denominators. Thus ϕ is the most poorly rationally approximable number. A system choosing ϕ as its frequency ratio avoids all low-order rational resonances and remains stable.

5.2 The Fibonacci Sequence as Integer Approximation to ϕ

$$F_0 = 0,\ F_1 = 1,\ F_{k+1} = F_k + F_{k-1}, \qquad \lim_{k\to\infty} \frac{F_{k+1}}{F_k} = \phi. \tag{11}$$

In FFGFT, the allowed recursion depths are:

$$\mathcal{N} \in \{F_k \mid k \in \mathbb{N}\}. \tag{12}$$

5.3 The Golden Angle $137.5°$

The golden angle:

$$\alpha_{\text{gold}} = 2\pi\left(1 - \frac{1}{\phi}\right) = 2\pi \cdot \phi^{-2} \approx 137.508°. \tag{13}$$

A growth system rotating successive elements by α_{gold} creates the densest possible packing without overlap — the reason for phyllotaxis patterns in sunflowers, pine cones, and leaf arrangements.

6 Level 5: Derived FFGFT Formulae

6.1 Fractal Coupling Constant $\alpha = \phi^{-1}$

$$\alpha = \phi^{-1} = \frac{2}{1+\sqrt{5}} \approx 0.6180, \qquad (x_{k+1}, y_{k+1}, z_{k+1}, ct_{k+1}) = \phi^{-1} \cdot (x_k, y_k, z_k, ct_k). \tag{14}$$

6.2 The $1/n^2$ Spectrum

The stable modes of the 4D torus have energies:

$$E_n \propto \frac{1}{n^2}. \tag{15}$$

Hydrogen, excitons, Rydberg atoms — all manifestations of the same spectrum.

6.3 Lepton Mass Ladder

$$\frac{m_\mu}{m_e} \approx \phi^5 \cdot \pi^2 \approx 206.7 \quad (\text{measured: } 206.77), \qquad \frac{m_\tau}{m_\mu} \approx \phi^5 \approx 16.94 \quad (\text{measured: } 16.8 \tag{16}$$

$$m_k \propto \phi^{-2k} \cdot m_0, \quad k = 0, 1, 2. \tag{17}$$

6.4 Fine Structure Constant α_{EM}

The fine structure constant follows from ξ and the characteristic energy scale E_0 (see Doc. 011 for the full derivation):

$$\alpha_{\text{EM}} = \xi \cdot \left(\frac{E_0}{1\,\text{MeV}}\right)^2 = \frac{4}{3} \times 10^{-4} \times (7.398)^2 = \frac{1}{137.04} \tag{18}$$

with $E_0 = \sqrt{m_e \cdot m_\mu} \approx 7.398\,\text{MeV}$.

Reference to Doc. 011

The full derivation — alternative formulations, historical context (Sommerfeld), and the fundamental power relation $\alpha \propto \xi^{11/2}$ — is documented in Doc. 011. This section reproduces only the key formula.

7 Self-Organisation Below the Wavelength Limit

7.1 The Physical Limit of Lithography

Modern chip manufacturing reaches a fundamental physical limit: structures can only be directly "written" with light if their dimensions exceed half the wavelength. Even with extreme ultraviolet lithography (EUVL, $\lambda \approx 13.5$ nm), structures below ~ 10 nm are no longer directly exposable [4, 5].

7.2 Directed Self-Assembly (DSA)

Directed Self-Assembly (DSA) of block copolymers (BCPs) provides the solution. A block copolymer consists of two chemically different polymer chains covalently bonded together. Since chemically dissimilar blocks repel each other, the chains spontaneously separate into periodic nano-phases — without every individual structure being externally programmed:

- **Periodicities:** 5–50 nm, far below the EUV wavelength.
- **Temperature:** activates phase separation and selects which mode (lamellar, cylindrical, spherical) is preferred.
- **Voltage/substrate:** guides modes into prescribed paths (guided DSA).

A concrete example: with a template pitch of 150 nm, DSA produced a hexagonal hole pattern with a pitch of 50 nm — a density tripling relative to the optically exposed pattern.

Decisive Observation

The material finds structures *smaller* than the wavelength of the incident light — not because the light creates these structures, but because the system spontaneously drifts into geometrically stable states. The external excitation scale and the resulting structure scale need not agree.

7.3 FFGFT Interpretation

The phase condition (1) has stable solutions at discrete mode numbers n — the $1/n^2$ spectrum. The block copolymer locks into the mode n^* that is energetically minimal given temperature T and substrate constraint U:

$$n^*(T, U) = \arg\min_n \big[E_n - \mu(T) \cdot n - \lambda(U) \cdot n^2\big]. \quad (19)$$

FFGFT: Sub-Wavelength Fixed Points

A system need not lock in at the scale of its excitation. The recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ has fixed points at all scales — the realised fixed point depends on boundary conditions, not on the excitation wavelength. Block copolymers under EUV illumination demonstrate: the system finds structures far below the light wavelength because the geometric condition (1) is scale-invariant.

7.4 Connection to the Form Hierarchy

The nano-patterns emerging from DSA — lamellae, cylinders, spheres, gyroids — are exactly the geometric basic forms from Section 2: $n = 2$ (lamellar), $n = 3$ (hexagonal-cylindrical), $n = 6$ (cubic-spherical). The material does not choose randomly — it solves the same phase condition that generates polygons, crystal lattices, and quantum numbers at every other scale.

8 Universal Manifestation in Nature

Table 1. Manifestation of the phase condition across scales

System	**Scale**	n **or** ϕ^{-k}	**Appearance**
Sub-Planckian	$\sim 10^{-48}$ s	$n = 1, 2, 3$	Fundamental structure ξ
Particles	10^{-18} m	$k = 0, 1, 2$	Lepton mass ladder
Atomic orbitals	10^{-10} m	$n = 1, 2, 3, \ldots$	Hydrogen spectrum
Molecules	10^{-9} m	$n = 6$	Benzene ring, DNA helix
Crystal lattices	10^{-7} m	$n = 3, 4, 6$	TFLN, graphene, honeycomb
DSA nano-patterns	10^{-8} m	$n = 2, 3, 6$	Chip manufacturing
Biology	10^{-3} m	ϕ^{-k}	Phyllotaxis, spirals
Galaxies	10^{21} m	ϕ-spiral	Spiral galaxies

Core Statement of FFGFT

None of these appearances are coincidences or analogies. All solve the same Eq. (1) at different n and at different scales. Self-organisation is geometrically necessary: every system drifts into the fixed points of the recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$, because no other stable states exist.

9 Conclusion

- A single equation — $\oint d\phi = 2\pi n$ — is the origin of all geometric forms and preferred ratios.
- Regular polygons, circle, ellipse, and spiral are different solution classes of the same condition.
- The golden ratio ϕ is the most stable incommensurable frequency ratio — mathematically necessary, not empirical.
- The Fibonacci sequence is the integer approximation to ϕ and defines the allowed recursion depths in FFGFT.
- All central FFGFT formulae follow directly from this geometric hierarchy.
- The same hierarchy appears at all scales — from sub-Planckian structure to spiral galaxies.
- DSA in chip manufacturing: the material finds structures below the EUV wavelength because the phase condition is scale-invariant.

Key Message

Nature is not diverse *despite* a single underlying structure. Nature is diverse *because* a single geometric condition has infinitely many solutions at infinitely many scales. FFGFT is the formalisation of this fact.

References

References

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